

Agenda

- This Week
 - Class 4 – August 19th (2:00– 3:00, 3:30 – 4:30)
 - Class 5 – August 20th (10:30 – 12:30)
 - Class 6 – August 20th (2:00 to 4:00)
 - Class 7 – August 21st (1:00 – 3:00)
- Starter Quiz
- Recap Last Week
- Student 1 – Banking Experience Talk
- Core Banking Platform Engineering :
 - Key Considerations
 - Platform Evolutions
 - Mapping of Lab Activity
 - Drivers for New Core Platforms Development
 - Challenges leading to high failure rates
- Student 2 – Banking Experience Talk
-

Note:

- Mobile Phones – Do not bring to class or Completely Switch off
- If students have any specific topic they want to discuss, share the topic and we can try to schedule

Starter Quiz

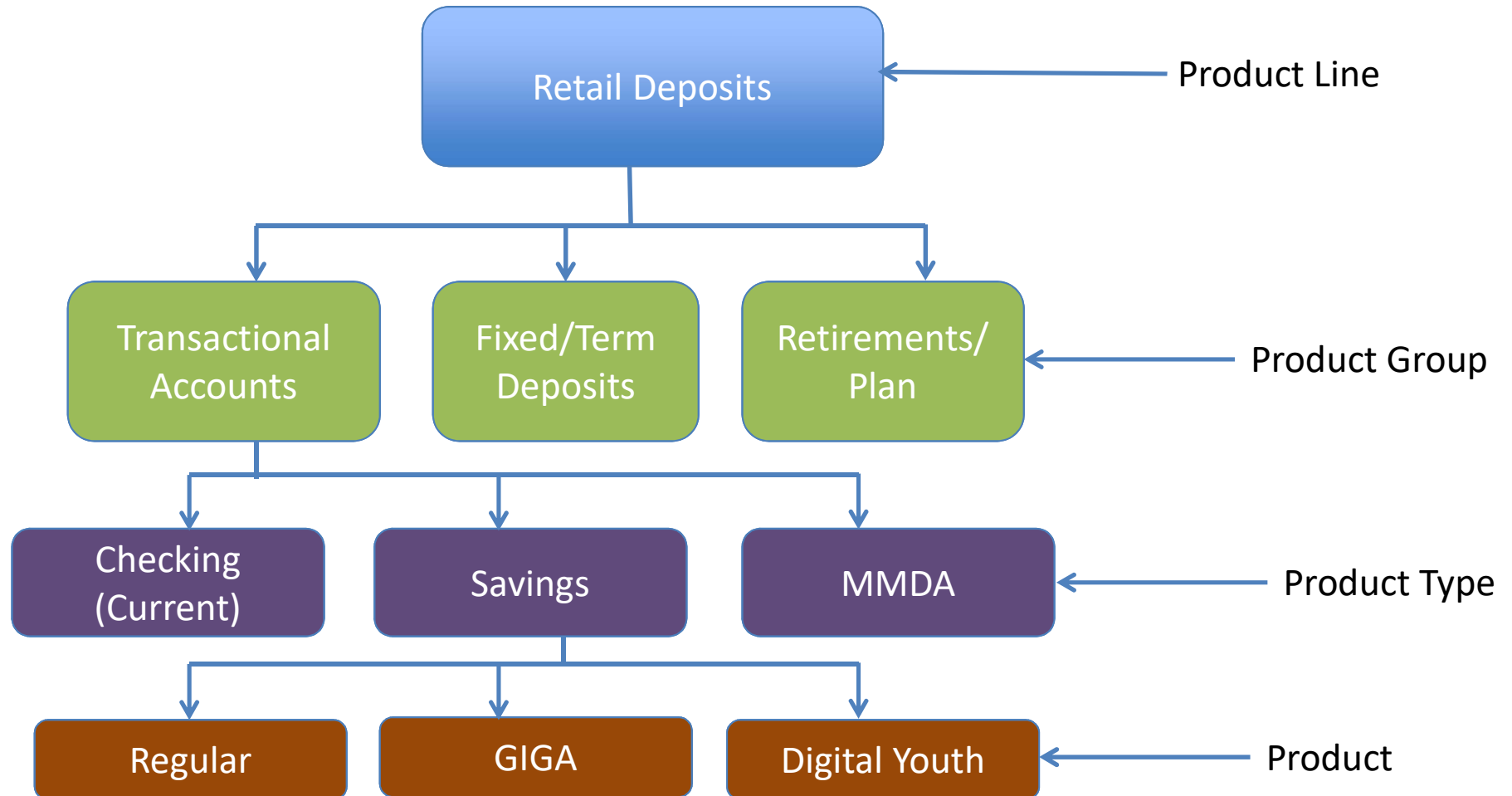
- Which Monthly Salary will U choose
 - 1 Crore Rupees Per Month
 - 1 Rupee Doubling every day for 30 days (Value on 30th day is your Salary Per Month – Do U Know much this would be ??)
- First Internet Banking Service in India – Year and Bank Name

Student 1 – Banking Experience Talk

- Two Student's per class for each day (August 19th, 20th morning and 20th evening, 21st) will be randomly selected (so a total of 8 for the week) and will have to come on stage and give a 15 Min talk of their Banking Product they use and details around the same such as
 - Bank Name
 - Bank Product ('s) Name and Type
 - ***What is the Format of your Bank Account Number (Size, Is there a Check Digit, Is it based on your Branch Code etc..)***
 - What Apps (Bank Channels, Payment Apps) do you use to operate and initiate Transactions on the Account
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 - What Interest and Fees are charged on your account and What is the Frequency of these Charges
 - What is the Interest Rate and When was the Last Interest Payment done by the Bank
 - What kind of notifications/communications specific to your account you receive from your Bank
 - What Improvements you would like to see in your Interaction experience with your Bank
 - Any thing else U would like to share on your Bank / your Banking Product

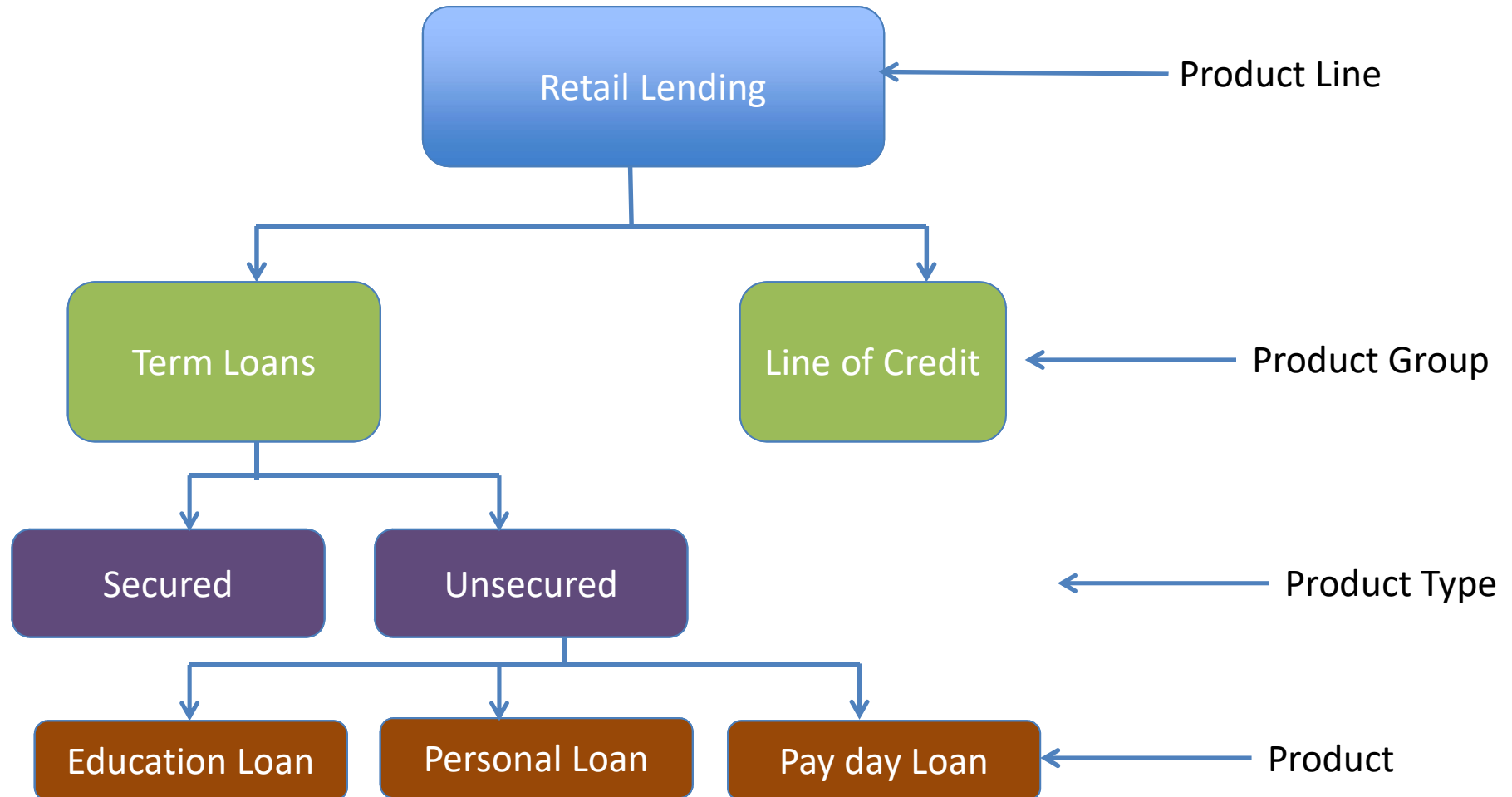
Retail Deposits Recap

Retail Deposits Sample Product Hierarchy



Retail Lending Recap

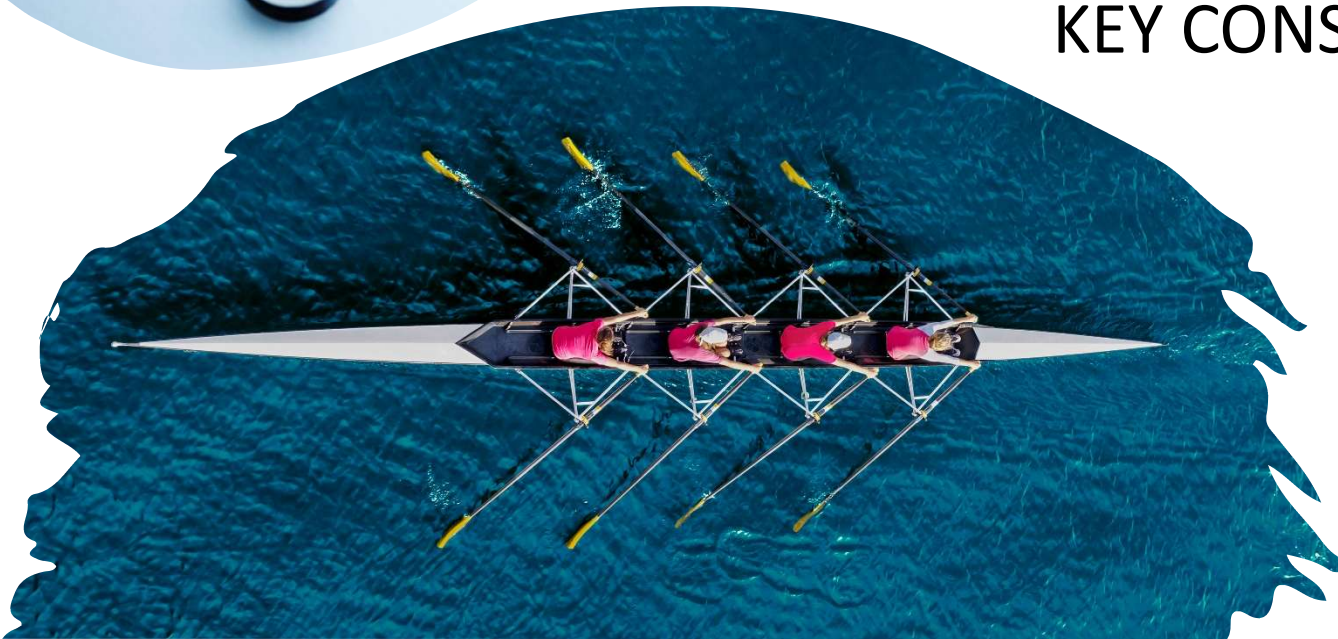
Retail Lending Sample Product Hierarchy





Core Banking Platform Engineering Fun and Challenging

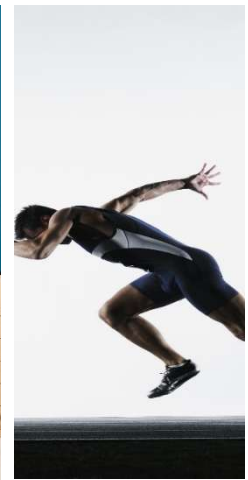
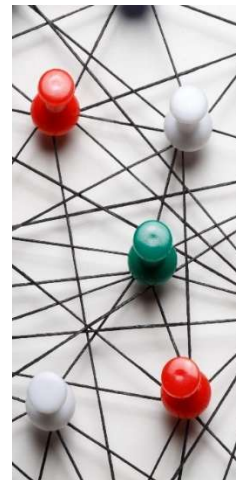
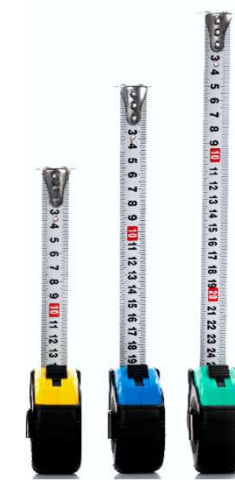
KEY CONSIDERATIONS



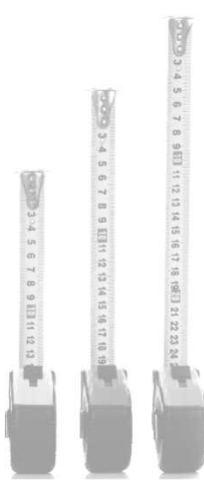
Core Banking Platform Engineering

Key Considerations

Core Banking Platform Engineering Key Considerations



Core Banking Platform Engineering Key Considerations



ROBUST SECURITY

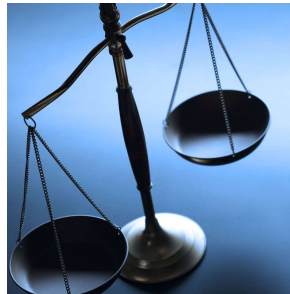
Impact of Security Breaches on Banks

Financial Losses and Legal Implications



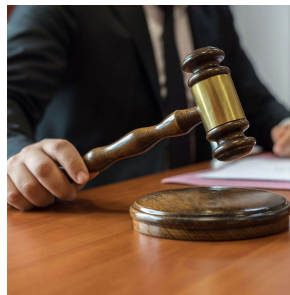
Direct Financial Losses

Breaches cause immediate monetary losses that affect the bank's financial health and operational stability.



Regulatory Fines

Violations trigger regulatory fines, increasing the cost burden and affecting the bank's compliance standing.



Legal Actions

Costly legal proceedings follow breaches, further draining resources and damaging reputation.

Reputation and Customer Trust

Impact of Trust Loss

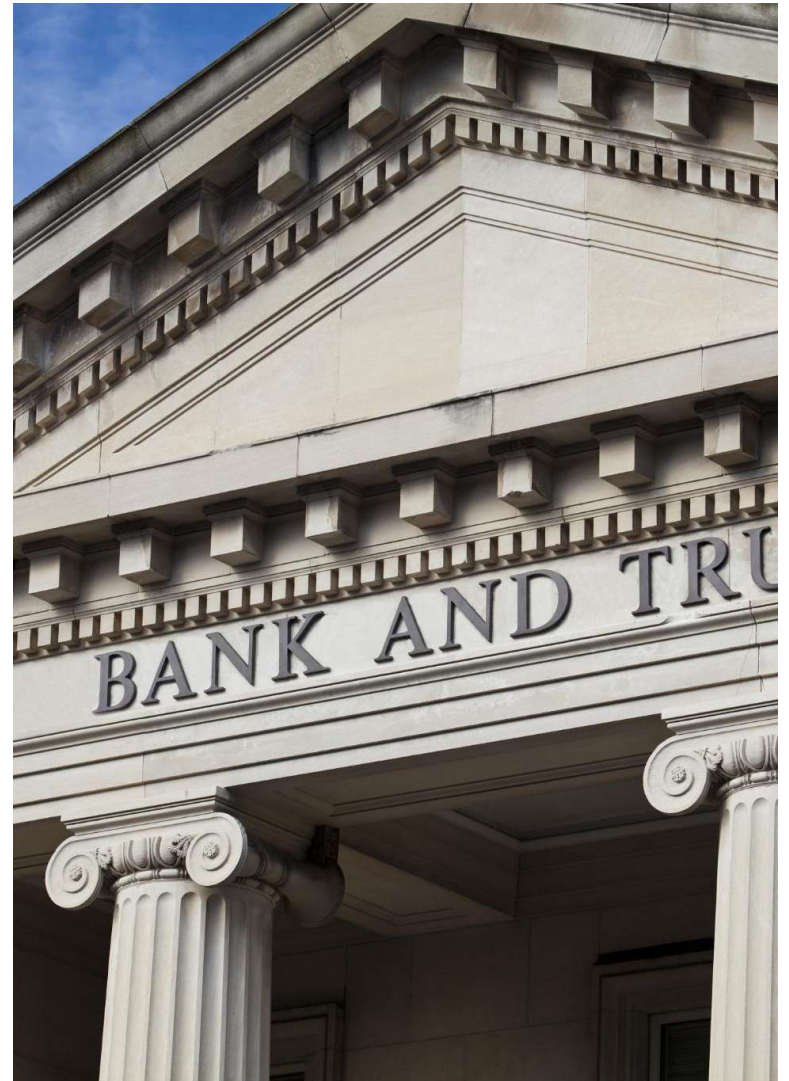
Loss of customer trust significantly damages a bank's reputation and long-term success.

Customer Retention Decline

Reduced trust leads to lower customer retention, affecting business stability.

Acquisition Challenges

Reputation damage makes acquiring new customers more difficult and costly.





Operational Disruptions

Impact on Operations

Security breaches interrupt daily banking functions, leading to unexpected operational downtime.

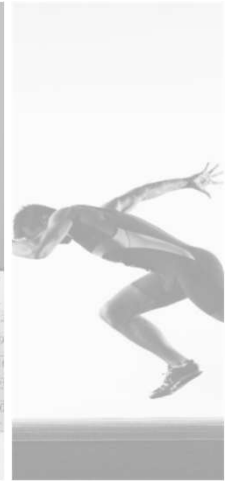
Transaction Delays

Disruptions cause delays in processing transactions affecting customer experience and trust.

Increased Recovery Costs

Recovering from security incidents results in higher financial and resource expenses for banks.

Core Banking Platform Engineering Key Considerations



REGULATION &
COMPLIANCE

Common Examples of Regulatory Non- Compliance in Core Banking



Money laundering and inadequate KYC procedures

Weak KYC Processes

Inadequate KYC procedures fail to properly identify customers, increasing risk of illegal activities.

Money Laundering Risk

Poor KYC controls enable money laundering by allowing suspicious transactions to proceed undetected.

Legal and Reputational Impact

Banks face significant legal penalties and damage to reputation due to weak KYC and money laundering.

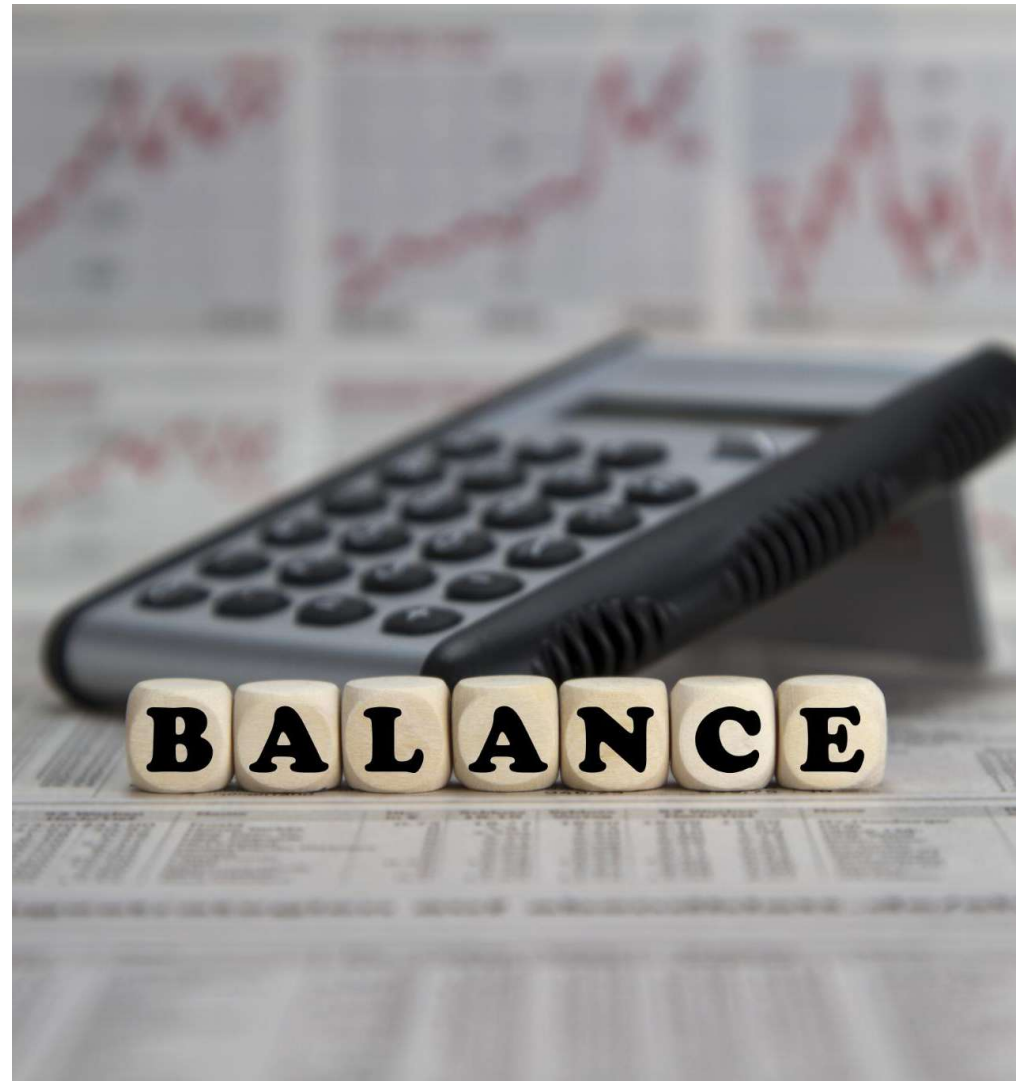
Failure to report suspicious transactions

Mandatory Reporting Duty

Banks must report suspicious transactions promptly to comply with regulatory requirements and prevent illegal activities.

Consequences of Non-Compliance

Failure to report suspicious transactions risks regulatory penalties and enables financial crimes to persist.

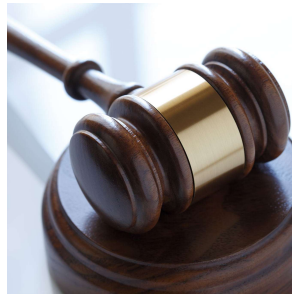


Non-adherence to data privacy regulations



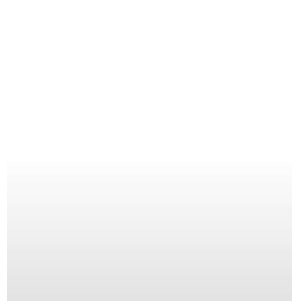
Importance of Data Protection

Protecting customer data is crucial to maintain privacy and prevent unauthorized access or breaches.



Consequences of Non-compliance

Failing to comply with data privacy laws can result in legal actions and financial penalties for organizations.



Impact on Customer Trust

Data breaches due to non-adherence damage customer trust and harm the bank's overall credibility.

Impacts of Non-Compliance on Financial Institutions



Legal and financial penalties

Consequences of Regulatory Breaches

Violating regulations leads to substantial fines and legal actions against institutions.

Financial Strain on Institutions

Fines and sanctions impose significant financial pressure, impacting profitability.

Reputational damage and loss of customer trust

Non-Compliance Consequences

Non-compliance causes significant reputational damage beyond financial penalties for institutions.

Customer Trust Erosion

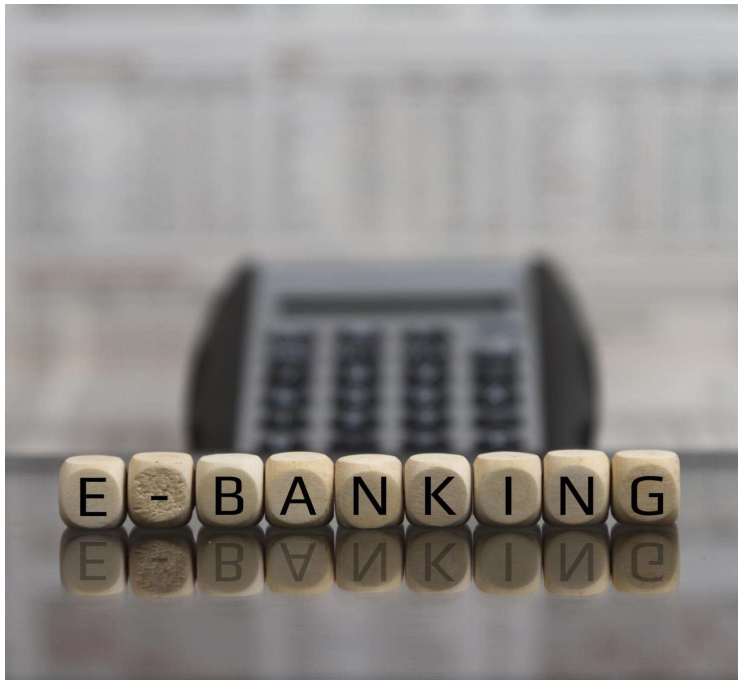
Loss of customer trust leads to clients switching to competitors, reducing loyalty and engagement.

Market Share Decline

Damaged reputation can shrink market share, hurting the institution's competitive position.



Operational disruptions and increased scrutiny



Regulatory Failures Impact

Failures in regulatory compliance cause significant disruptions in daily banking operations.

Increased Regulatory Monitoring

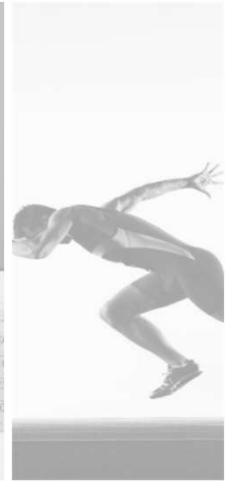
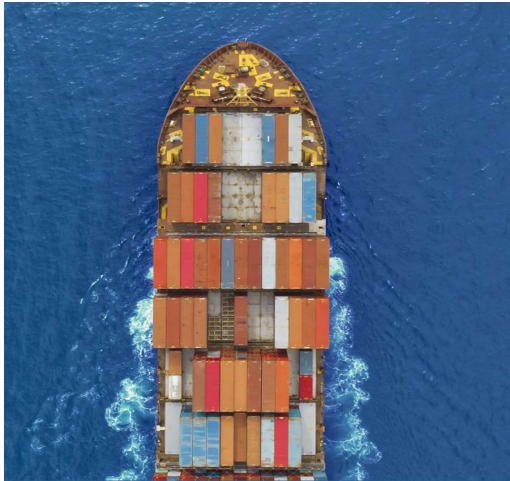
Regulatory failures result in heightened scrutiny and continuous oversight from regulators.

Resource-Intensive Corrective Actions

Addressing failures requires resource-heavy corrective measures affecting bank efficiency.

Core Banking Platform Engineering

Key Considerations



LARGE DATA
VOLUMES

Large Data Volumes - Insights and Operational Implications

Cross-Country Data Volume Comparison

Data Volume Leadership

Indian and Chinese banks have the highest account and transaction volumes influenced by the country's large population.

Comparative Bank Data

Indian and US banks follow China in data volumes reflecting different population sizes and banking usage.

Customer Behavior Impact

Variations in customer behavior and banking models shape unique data patterns across countries.



Operational Challenges at Scale

Latency Management

Minimizing delays in data processing is critical to maintain efficient system performance at scale.

Data Consistency

Ensuring all data remains accurate and synchronized across distributed systems is essential for reliability.

Security and Compliance

Protecting data privacy while meeting regulatory standards requires robust security measures and processes.

System Availability

Maintaining continuous access to data and services demands resilient infrastructure and real-time monitoring.



Strategies for Managing Large-Scale Data in Core Banking

Cloud Computing Adoption

Banks use cloud computing to scale infrastructure dynamically and manage large volumes of data efficiently.

Microservices Architecture

Microservices enable flexible and modular banking applications to improve system resilience and agility.

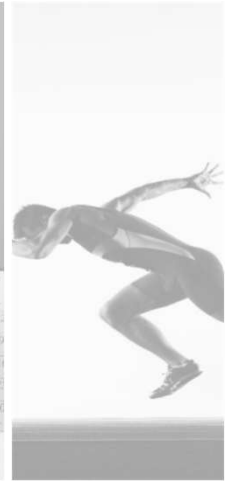
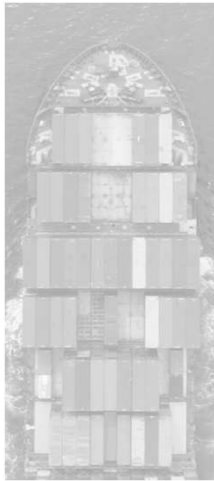
AI-Powered Analytics

Artificial intelligence analytics help banks detect patterns, predict trends, and improve decision-making.

Real-Time Monitoring

Continuous monitoring ensures data integrity and optimizes system performance in real-time.

Core Banking Platform Engineering Key Considerations



HIGH SCALABILITY &
ELASTICITY

Large Chinese, Indian, American Banks: Transaction Volume Insights

Analysis of daily and annual transaction volumes

High Transaction Volumes

Large Chinese, Indian, American banks experience daily transaction volumes exceeding Tens of Millions to Hundreds of Millions due to vast user activity.

Large Population Impact

India and China's large population significantly contributes to the massive scale of banking transactions.

Rapid Digitization

Rapid adoption of mobile payment platforms supports extensive transaction processing in India and China.





Factors influencing transaction volumes

Mobile Payments Adoption

Widespread use of mobile payments significantly increases transaction volumes in Chinese banks.

Government Financial Inclusion

Government initiatives promote financial inclusion, expanding banking transaction activity across populations.

Urbanization Impact

Rapid urbanization drives demand for banking services, increasing transaction volumes in cities.

Digital Finance Innovation

Technological innovations in digital finance enable seamless transactions, boosting volume and efficiency.

Future outlook for core banking transaction volumes globally



Growth Drivers

Digital finance expansion and rising customer expectations drive steady growth in banking transactions worldwide.

Scalability Needs

Banks must enhance systems to handle increased transaction volumes efficiently and reliably.

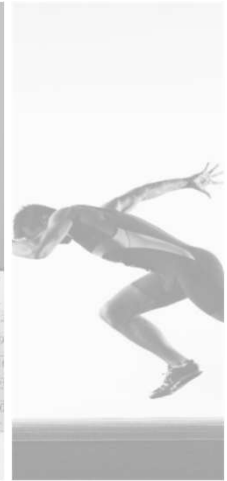
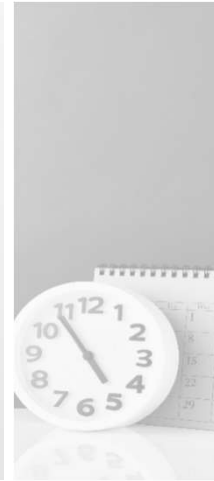
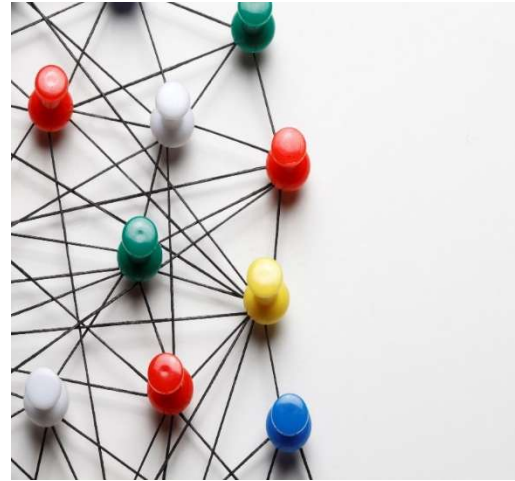
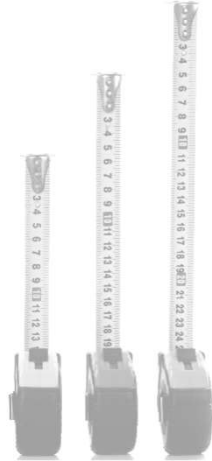
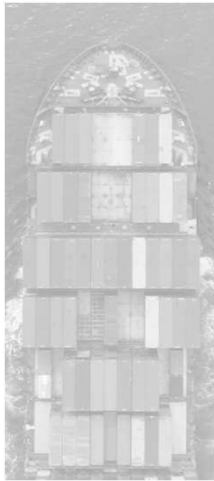
Security Focus

Heightened security measures are critical to protect transactions and customer data against evolving threats.

Customer-Centric Innovation

Innovations that improve customer experience are key to staying competitive in the evolving banking sector.

Core Banking Platform Engineering Key Considerations



LARGE INTEGRATION
ECOSYSTEM

Corebanking Platform Ecosystem Components

Bank Channel Apps



Payment & Embedded Finance Apps

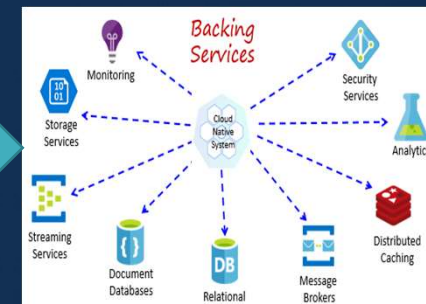


**Core Banking
Platform
(Modular Core
Product
Processors)**

Enterprise Components

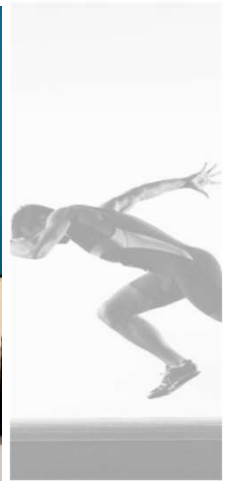
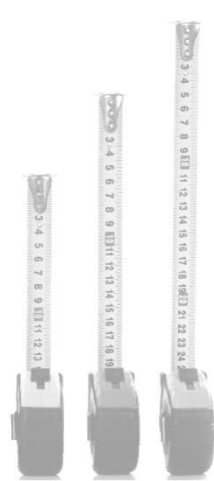
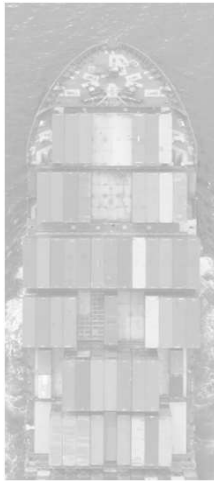
- Customer
- Bank Definition
- General Ledger
- Alerts Rendering
- Payments Hub
- ...

Enterprise/Cloud Services



Core Banking Platform Engineering

Key Considerations



HIGH AVAILABILITY
24*7*365

High Availability in Core Banking

What constitutes high availability

Continuous Operation

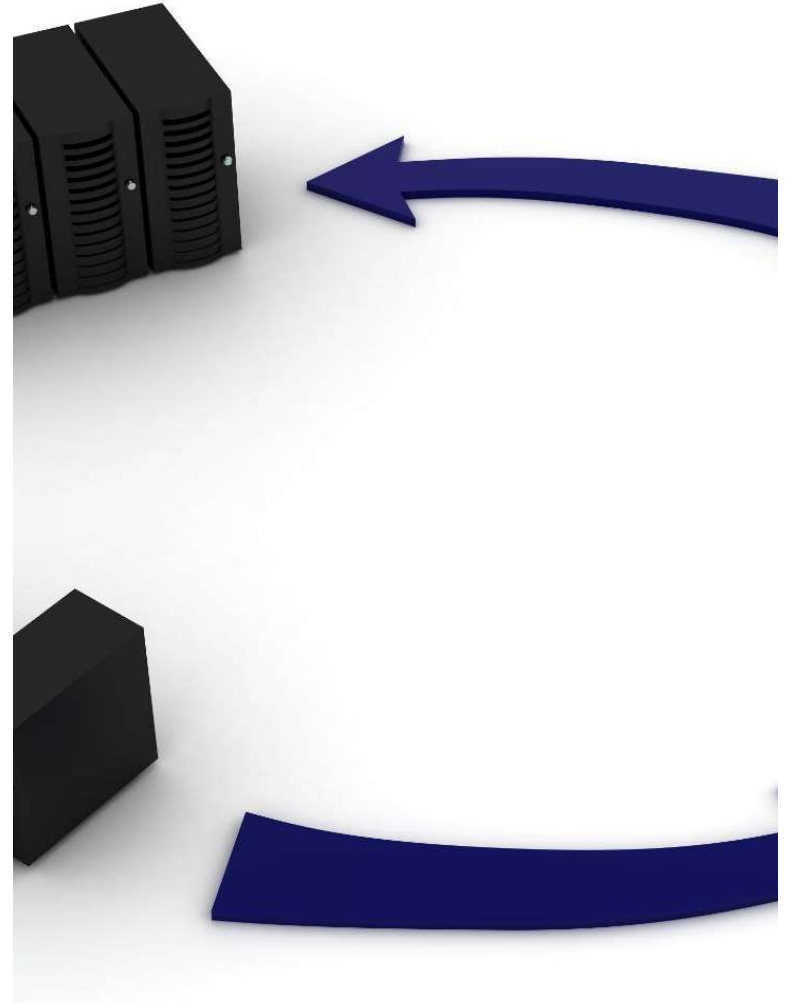
Systems are designed to operate without failure for long periods, ensuring ongoing service availability.

System Redundancy

Incorporating redundancy ensures backup components can take over instantly if primary ones fail.

Rapid Recovery

Quick recovery mechanisms minimize service interruptions and restore functionality promptly after failures.



Measuring and monitoring availability



Uptime Percentage

Uptime percentage measures the total time a system is operational and available for use.

Mean Time Between Failures

MTBF indicates the average time interval between system failures, reflecting reliability.

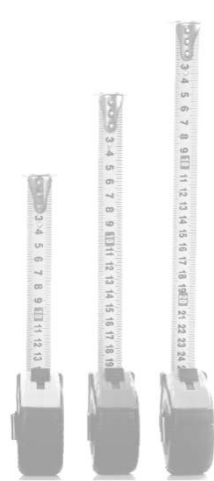
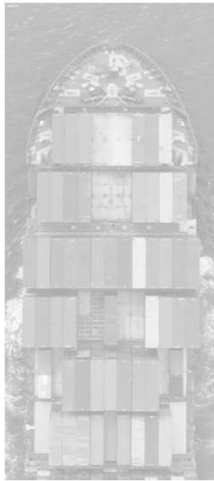
Mean Time to Recovery

MTTR measures the average time taken to recover and restore system functionality after failure.

Continuous Monitoring Tools

Continuous monitoring tools support real-time tracking of availability and system performance metrics.

Core Banking Platform Engineering Key Considerations



**AGILITY WITH
CERTAINTY**

Student 2 – Banking Experience Talk

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Corebanking Platform Evolution



Core Banking Platform Evolution – GEN 1,2,3

Gen 1 Core

AGE: > 40 years
TECHNOLOGY: MAINFRAME
ARCHITECTURE: MONOLITHIC
LANGUAGE: COBOL/PL1/
ASSEMBLER
DATABASE: VSAM/DB2
HOSTING: ON-PREM,
DATACENTER

Gen 2 Core

Age > 25 years
Technology Open Systems (Unix)
Architecture Largely Monolithic
Languages 4GLs, C
Database Relational DB / Prop
Hosting On-Prem, Datacenter

Gen 3 Core

Age <=7 years
Technology Cloud Technologies (Docker/ Kubernetes)
Architecture Component Based, 100% API & Event Enabled, Full Automation, Highly Extensible
Languages Java / Python
Databases Relational DB and Big Data
Hosting Public or Private Clouds

- Core Code Customization done heavily in Gen 1 and Gen 2 Core's where as Gen 3 Core's are designed with High Extensibility with out modifying Core Code
- Customer Segmentation in Gen 1 and Gen 2 Core – Specific Segments (like Large Banks or Smaller Digital Only Banks ..) where as Gen 3 Cores can support multiple Customer Segments

Note: In you Lab project, use as many as Gen 3 Core Technologies as possible

Core Banking Platform Evolution – Monolithic to Micro Services

Structure and Characteristics of Monolithic Systems

Tightly Coupled Components

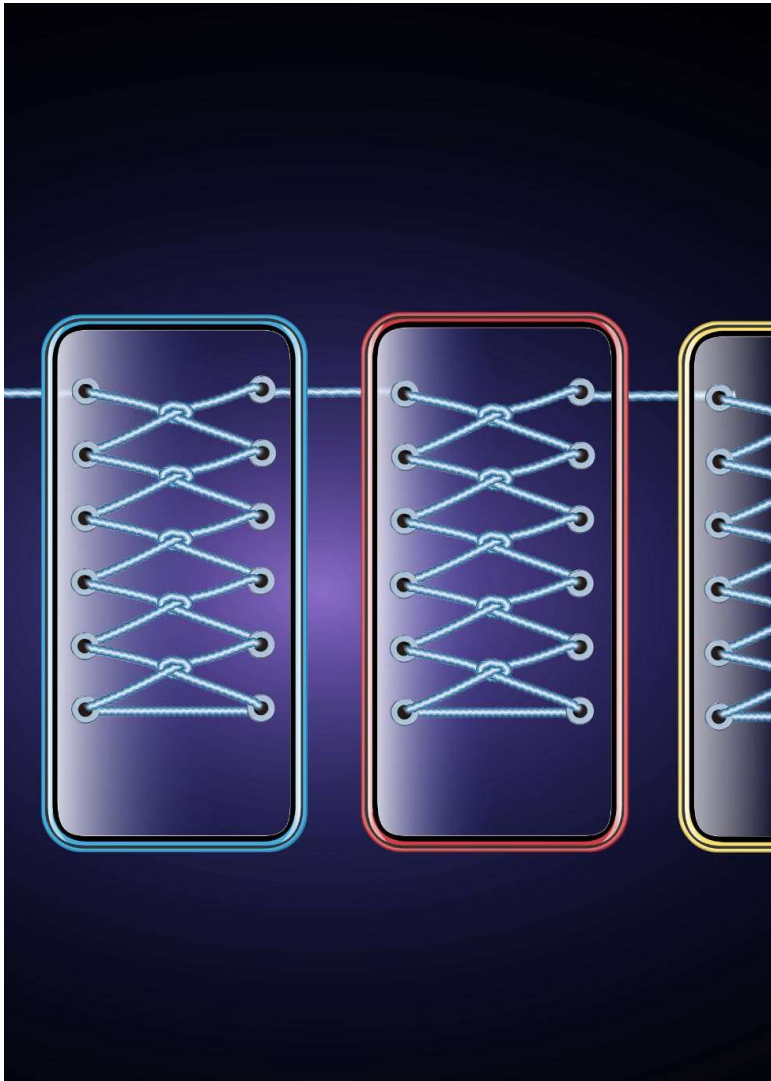
Monolithic systems have tightly coupled components that depend heavily on each other, limiting modularity and flexibility.

Single Codebase

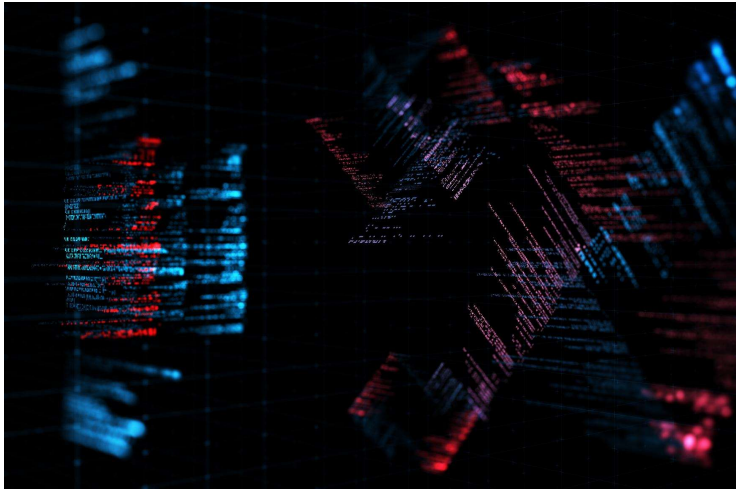
All features and functions reside in a single codebase, which simplifies development but complicates scalability.

Single Deployment

Deployments affect the entire application, which can reduce agility and increase risk during updates or changes.



Core Banking Platform Evolution – Monolithic to Micro Services



Limitations and challenges of monolithic cores

Limited Scalability

Monolithic cores struggle to scale efficiently, limiting system growth and performance improvements.

Complex Maintenance

Large monolithic codebases are difficult to maintain and evolve, increasing the risk of errors.

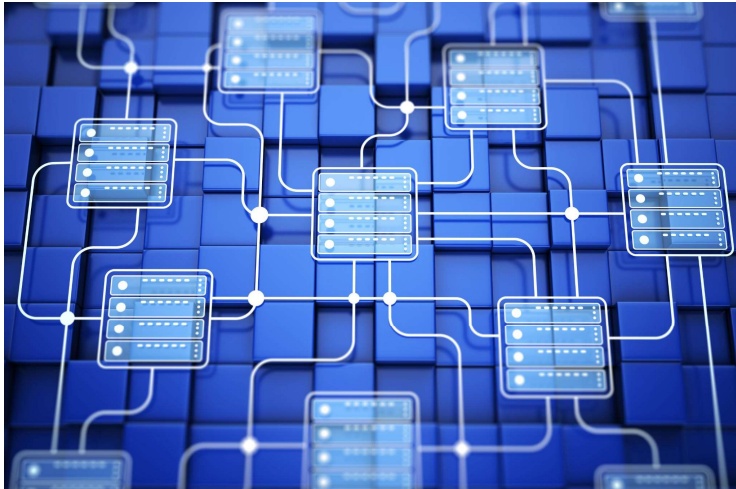
Slower Innovation

Implementing changes in monolithic systems is challenging, slowing down the pace of innovation.

Risk of System Failures

Failures in monolithic cores can affect the entire system, increasing downtime risks.

Core Banking Platform Evolution – Monolithic to Micro Services



Overview and Key Principles of Microservices

Modularity

Microservices break down applications into modular, independent components each with its own small independent Services for easier management and development.

Independent Development and Deployment

Each microservice can be developed and deployed independently, allowing faster updates without impacting the entire system.

Scalability

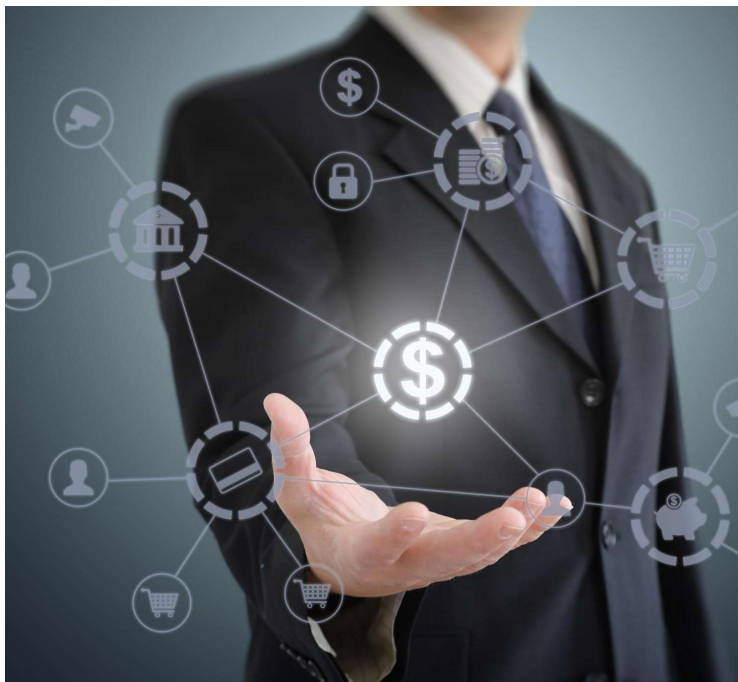
Microservices architecture enables scalable systems by allowing individual components to scale based on demand.

Continuous Delivery

Continuous delivery ensures rapid, reliable software releases, enhancing agility and innovation in banking technology.

Core Banking Platform Evolution – Monolithic to Micro Services

Advantages of Micro Services Architecture



Improved Scalability and Flexibility

Core banking systems benefit from enhanced scalability and flexibility, supporting growing business demands effectively.

Faster Development Cycles

Faster development cycles enable banks to deploy new features quickly and respond to market changes.

Fault Isolation and Integration

Fault isolation improves system stability while easier integration allows seamless adoption of new technologies.

Agile Response to Needs

Supports agile adaptation to changing customer demands and regulatory requirements in the banking sector.

Core Banking Platform Evolution – Monolithic to Micro Services



Common challenges and considerations

Architectural Complexity

Adopting new systems increases architectural complexity, requiring careful design and integration strategies.

Inter-Service Communication

Managing communication between services is crucial for system reliability and performance.

Data Consistency

Ensuring data consistency across distributed components poses significant challenges.

Security and Monitoring

Robust security measures and monitoring tools are essential to protect and maintain the system.

Core Banking Platform Evolution – Monolithic to Micro Services

Examples of Microservices Implementation in Banks

Modular Core Product Processors (e.g. Retail Deposits Transactional Account Core Product Processor, Fixed Deposits Core Product Processor etc..)

Microservices enable scalable and efficient Account Management and Transaction Processing in modern banking environments.

Customer Onboarding and Customer Data Management

Microservices streamline customer onboarding, enhancing user experience and operational responsiveness.

Fraud Detection

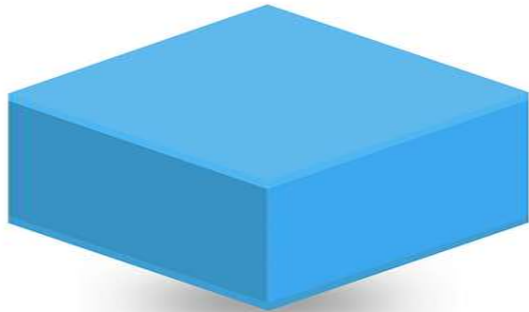
Fraud detection microservices improve security by rapidly identifying suspicious banking activities.



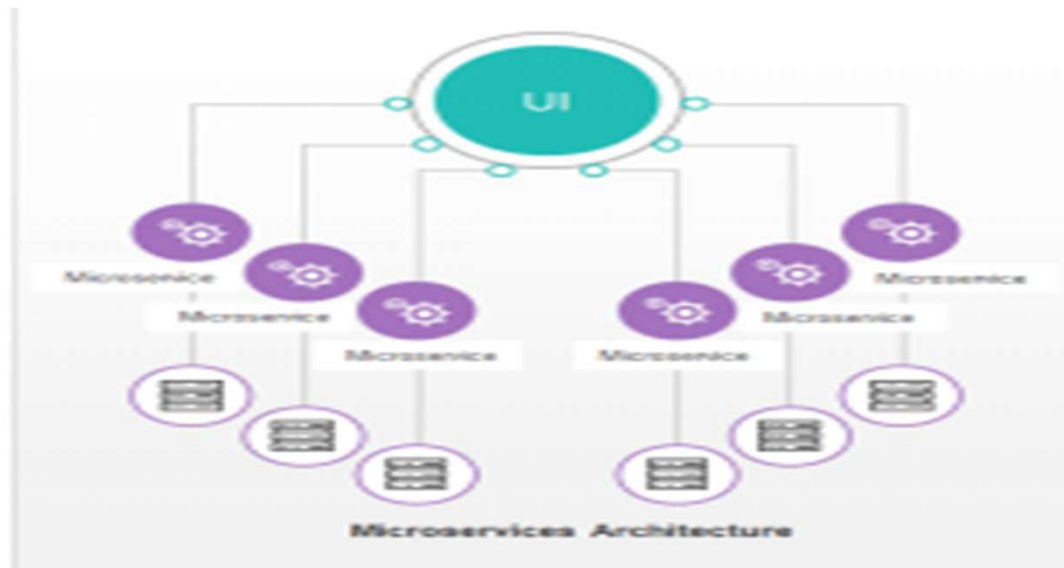
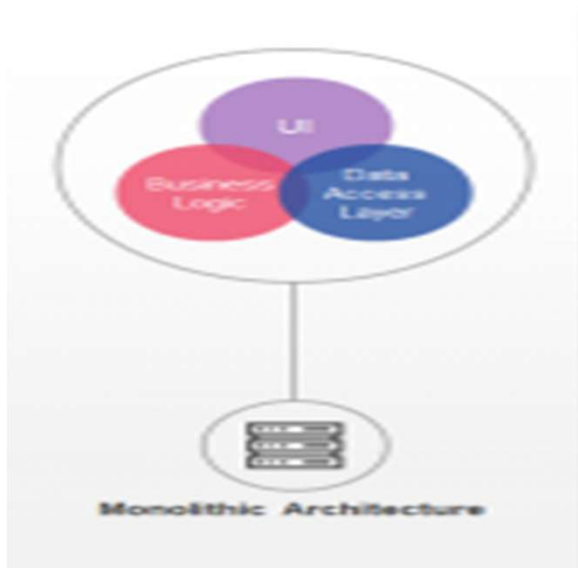
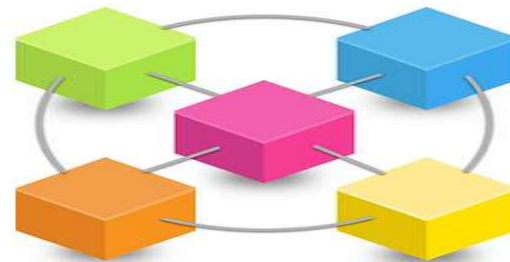
Comparative Analysis: Monolithic vs Microservices

Core Banking Platform Evolution – Monolithic to Micro Services

Monolithic

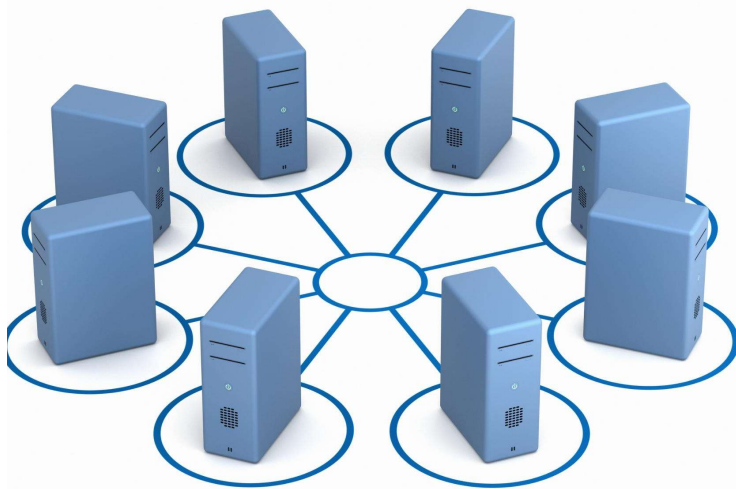


Microservices



Core Banking Platform Evolution – Monolithic to Micro Services

Scalability and Flexibility Considerations



Independent Component Infrastructure

Base Infrastructures that can provide optimal value can be leverage based on the nature of the Component Processing (e.g. Low Coast Database like Postgres can be used for simple CRUD Components and a Higher cost Database like Oracle can be used for High Transaction Processing Components)

Independent Component Scaling

Microservices architecture enables each component to scale independently, improving resource efficiency and responsiveness.

Monolithic System Limitations

Monolithic systems scale as a whole, which can lead to inefficiencies and reduced flexibility in resource management.

Core Banking Platform Evolution – Monolithic to Micro Services

Impact on Maintenance, Upgrades, and Innovation

Microservices Enable Independent Upgrades

Microservices allow independent deployment of components, speeding up upgrades and reducing maintenance downtime.

Monolithic Systems Require Full Redeployment

Monolithic architectures often need complete redeployment for updates, causing longer maintenance periods and reduced agility.

Innovation Through Service Isolation

Isolated microservices promote faster innovation by enabling teams to develop and deploy features independently.



Let us Map Your Lab Activity with Gen 3 Core Attributes

- What is the Solution you are building
 - Simple Retail Deposits – Term Deposits Core
- What Components in the Solution are you building consuming Cloud Native Services
 - Customer
 - Retail Deposits – Term Deposits Core Product Processor
- What Apps are building in each Component
 - Customer Component
 - UI App
 - API App
 - Event App
 - Batch/Bulk Processing App
 - Retail Deposits – Term Deposits Core Product Processor
 - UI App
 - API App
 - Event App
 - Batch/Bulk Processing App
 - Reporting App

Let us Map Your Lab Activity with Gen 3 Core Attributes

- What are some Key Technologies being used
 - Programming Languages:
 - Java – API, Event, Batch
 - Python – Reporting
 - Cloud Technologies
 - Docker
 - Kubernetes
 - CI-CD Pipeline
 - Jenkins
 - Cloud Services – Azure
 - Container Management (AKS)
 - Event (Azure Event Service)
 - Monitoring (Azure Monitoring Service)
 - Scheduler (Airflow)
 - Database Service (Postgres)

Let us Map Your Lab Activity with Gen 3 Core Attributes

- Some Pointers / Hints
 - Design, develop, deploy each of these Apps as separate Apps and Integrate them using Published Interfaces (Like API's, Events ..)
 - Deploy the Apps with an automated CI-CD Pipe Line
 - Need Environments for Component Development and also an Integrated Environment for Deploying both the components and all Apps of these components and certifying end to end
 - Collaboration and Communication within a Group and Across the Various Groups in a Team is an important element and learning of how actual work happens in real life
 - Usage of Cloud Monitoring Services to Monitor each App and all Apps in the "Solution"
 - Each of the App's should be designed to be Independently Deployable and Scalable
 - Testing Should be done at each Interface Level, App Level, Component Level and Solution Level

Drivers Of New Core Banking Platform Development

Core Banking Platform Evolution – Next Generation (Gen 4) Core Platform Thoughts

- Key Characteristics
 - Intelligent (using AI Technologies) – Some Use Cases
 - Product Offering and Configuration
 - Operations – Dynamic Infra Provisioning
 - ...
 - Ultra Scalability (Data and Transaction Volumes)
 - India UPI Volume Growth from 19.47 Billion in July 2025 to 54 Billion in 2030, Globally the same is being projected
 - Large Banks scaling to 200+M Transaction Accounts
 - High Elasticity
 - Large Spikes / Changes in Transaction Volumes
 - Continuous Evolution and Rapid Change Enablement
 - Rapid New Feature Rollout (using AI Tooling)
 - High Extensibility (Enabling Bank Engineers, Partners, 3rd Party Developers)
 - High automated CI-CD pipelines
 - + All of Gen 3 Core Features Like
 - Modular (i.e. Modular Core Product Processors)
 - Supporting all Integration Patterns
 - Realtime (API), Near Realtime (Event), Batch/Bulk (Files)
 - Private and Public Cloud Deployment support

Technological advancements and digital demands

Real time Transaction Growth (Incl Embedded Finance)

Real time Transaction (Incl Embedded Finance) Volume has surged, requiring banks to have Highly Scalable, Resilient, Performant and Real Time Core Banking Platforms

Digital / Mobile Apps Explosion

Digital / Mobile apps enable customers to manage finances securely and efficiently anytime, anywhere.

Fintech Influence

Fintech innovations drive banks to upgrade systems to enhance speed, security, and user experience.





Regulatory and compliance pressures

Increasing Regulatory Demands

Banks are subject to growing regulatory requirements and are subjected to heavy penalties for noncompliance, this is to ensure financial stability and consumer protection.

- TD Bank has been slapped with a record \$3 billion penalty for violations of the Bank Secrecy Act (BSA), the primary U.S. anti-money laundering law in 2024. A Number of Executive Leadership changes were done due to this.

Need for Robust Systems

Core banking systems must be strong, secure, and adaptable to meet evolving regulatory standards.

Audit and Risk Reduction

Auditable systems help banks reduce operational risks and maintain compliance with regulatory bodies.



Competitive landscape and customer expectations

Intense Competition

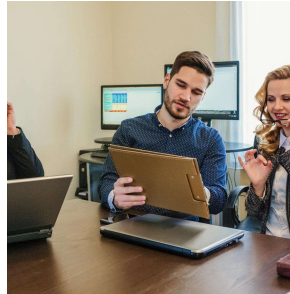
Banks face intense competition requiring continuous innovation to maintain market position and customer loyalty.

Rising Customer Demands

Customers increasingly expect personalized, real-time services that enhance convenience and satisfaction.

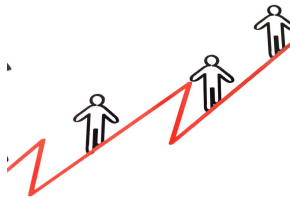
Challenges Leading To High Failure Rates

Lack of skilled resources, High Complexity and Quality Issues



Specialized Skills Scarcity

Core banking domain Engineering demands deep domain specialized skills and ground up build culture that are often difficult to find in the workforce.



High Complexity

Vast Functional Domain

Engineering Mission Critical, Always On, Highly secure, scalable, resilient platform with full Regulatory compliance and global adaptability/extensibility is incredible challenging



Quality Issues

Skill gaps and complex business domain with multitude of use cases contribute to quality problems

Complexity of legacy system migration



Data Volume Challenges

Large volumes of data complicate the migration process and require careful management to ensure integrity.



System Dependencies

Interdependencies between legacy systems increase migration complexity and risk of failures during transition.



Risk of Disruption

Migration carries high risk of business disruption, requiring stringent planning and risk mitigation strategies.

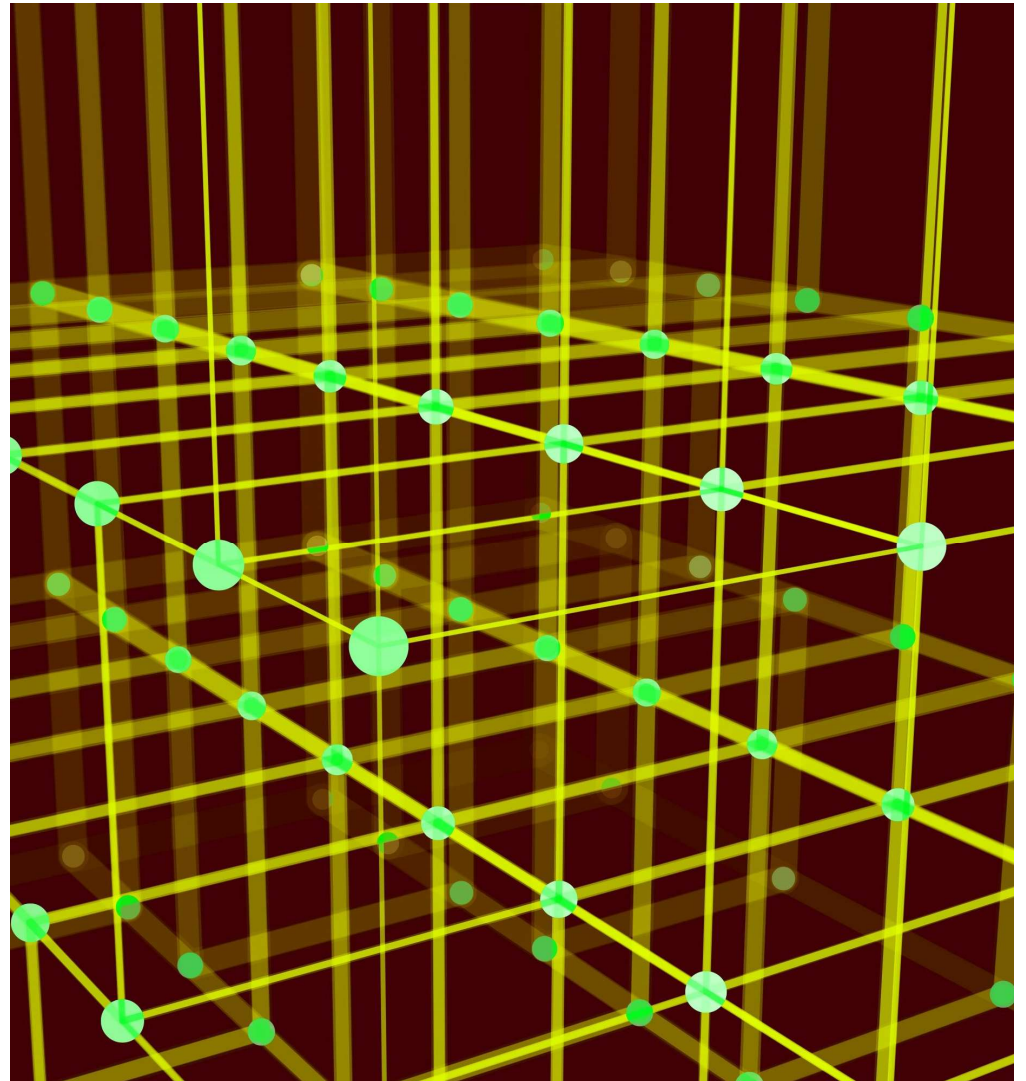
Integration difficulties with existing infrastructure

Seamless Platform Integration

New platforms need smooth integration with existing applications and third-party services to function effectively.

Incompatibility Issues

Technical gaps and incompatibilities between systems often cause delays or failures in integration projects.



Session Summary

